Towards Ethical and Personalized Web Navigation Agents: A Framework for User-Aligned Task Execution

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Abstract

Generative AI has advanced the capabilities of autonomous agents, enabling autonomous execution of complex web navigation tasks that can reshape digital interactions across various domains. Yet, to reach their full potential, these agents must be ethically aligned and personalized to individual user needs—a challenge complicated by privacy concerns and the risk of reinforcing biases. This work introduces a novel framework that enables responsible, user-guided personalization of web navigation agents, ensuring alignment with ethical standards and user preferences. By developing agents capable of perceiving, reasoning, and adapting in alignment with user preferences, this work proposes an approach that transcends generic task execution. Employing a structured representation of user-specific tasks, the agent utilizes interactive and reasoning actions to personalize workflows, adapting responsively to individual contexts. Evaluation through task success metrics and user satisfaction scores further assesses the ethical alignment and utility of personalized interactions. This research lays the groundwork for responsible agents that offer personalized assistance while adhering to ethical and privacy standards, with implications for information retrieval, e-commerce, and other knowledge-intensive applications.

CCS Concepts

• **Information systems → World Wide Web**.

Keywords

Web Navigation Agents, Intelligent Agents, Personalization

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1 Introduction

The rapid evolution of artificial intelligence (AI) is reshaping technology for complex problem-solving. Generative AI, in particular, enables intelligent agents to autonomously navigate challenges, adapt to new contexts, and meet diverse user needs. These advances promise a new era of digital interaction, where AI-driven agents replace rigid rule-based systems with flexible, goal-oriented decision-making.

Agents are autonomous entities capable of perceiving their environment, making decisions, and executing actions to achieve specific goals or maximize rewards. Traditional agentic systems had limited capabilities for natural language interaction and often relied on predefined rules or simple decision-making processes. Today, these challenges have largely been addressed by generative models, and multiple methods demonstrate promising results by utilizing LLMs as the foundation of autonomous agents [1, 8–12, 15, 18, 19].

Web navigation agents, capable of interacting with web environments through actions like `click` and `scroll`, can autonomously perform a wide range of complex tasks. In a world where life-altering actions, such as submitting job applications or making large financial purchases, can be completed with a few clicks, generalist web agents hold significant power, utility, and potential risk.

However, with great potential come equally significant concerns around safety, alignment, and trustworthiness. The introduction of generative AI-backed agents raises questions about their ability to consistently align with human goals and values, as well as their safety in environments that require nuanced judgment and decision-making. These concerns necessitate a deeper investigation into how agents handle issues like perception, planning, and execution, and what strategies can mitigate the risks involved in their deployment. Beyond examining existing evidence, there is also a critical need to identify gaps in our current understanding and establish directions for future research to ensure agents operate reliably and ethically.

2 Background and Related Work

Web navigation agents significantly reduce the time and effort users spend on intricate online activities by automating multi-step processes. Techniques like reinforcement learning and imitation learning enable them to navigate through webpages autonomously—locating products, selecting options, and completing purchases [15]. They achieve these tasks much faster than human users, who might spend several minutes navigating, whereas agents can perform similar actions in seconds [15].

Traditional web interfaces can overwhelm users with numerous options, contributing to decision fatigue. Web navigation agents alleviate this by curating information and limiting the number of choices shown, supporting a more streamlined decision-making experience. Excessive choices have been shown to lead to cognitive overload, reducing user satisfaction [13]. By presenting a filtered set of options, agents make it easier for users to make decisions without facing an overwhelming array of choices.

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These agents are especially useful for users handling unfamiliar tasks, such as managing financial accounts, navigating legal processes, or understanding health-related information. Web navigation agents can guide users through each step, explaining technical terms and automating repetitive actions [11]. While generative chatbots have demonstrated similar support in assisting users with complex tasks [17], web navigation agents offer additional structure by directly interacting with interfaces to assist users in real-time.

Web navigation agents also improve accessibility for users with disabilities, limited technical skills, or low digital literacy by simplifying online interactions. For instance, voice-command capabilities can assist visually impaired users in filling out online forms, navigating through complex layouts, and interpreting on-screen information [11, 14]. This adaptability allows a wider audience to benefit from online resources, making the internet more accessible.

3 Research Progress

My dissertation focuses on developing responsible generative agents for knowledge-intensive information retrieval (IR) tasks, aiming to build systems that can navigate complex information landscapes and synthesize knowledge across diverse sources. My research centers on three main areas: generative model applications, knowledge-intensive IR tasks, and alignment and evaluation methods. Together, these areas support the creation of agents capable of managing intricate workflows, which are essential for effective web agents.

In generative model applications, I have applied large language and vision-language models to tasks such as claim verification, data augmentation, and dialogue generation. For example, [6] leverages LLMs with knowledge graphs for verifying claims, while [5] explores dialogue generation. My research in [4, 7] uses vision-language models to generate synthetic data that address issues like weak decision boundaries. While these projects offer valuable insights into generative capabilities, they primarily serve as foundational elements rather than fully autonomous agent systems.

In knowledge-intensive IR tasks, my work includes creating an interpretable scoring system to evaluate LLM-generated answers based on knowledge, structure, and content [2]. This system validates answer reliability by comparing responses to knowledge graph data and aligning structure with user intent. Such a scoring mechanism is essential for web navigation agents, which rely on providing accurate, dependable information for complex tasks.

My focus on alignment and evaluation aims to ensure agents act responsibly, particularly in open-ended environments. In [5], I introduced an evaluation framework to detect covert harms in conversational agents. My current efforts involve developing a knowledge evaluation dataset for LLMs using knowledge graph triplets. These frameworks help align agents ethically and ensure responsible operation in autonomous settings.

The work presented in [3] demonstrates an application of web navigation agents in e-commerce. The agent autonomously identifies subjective product attributes, engages in conversations to clarify user preferences, and generates personalized gifting ideas. By minimizing cognitive effort and enhancing product discovery, this project highlights the potential of agents to effectively handle subjective and nuanced tasks.

Together, these projects advance my goal of building web navigation agents that integrate knowledge synthesis, task automation, and ethical alignment, providing high-quality, responsible assistance across diverse applications.

4 Motivation and Research Questions

Web navigation agents offer tremendous versatility, with the potential to perform a wide array of tasks across domains since most activities can be accomplished online through a combination of clicks and keystrokes. However, to achieve true utility, these agents must be sufficiently personalized – not only regarding *what* tasks are executed but also *how* they are performed. For these agents to effectively assist users, they need to align closely with each user's unique preferences, values, thought patterns, and behaviors.

A significant challenge lies in acquiring the necessary personalization data in an ethical manner. Ensuring that users are fully aware and consenting participants in the data collection process is vital. Traditional passive tracking and behavioral analysis approaches, which often operate without explicit user involvement, may be perceived as intrusive or ethically problematic. Therefore, a new framework is required that enables transparent, user-guided data collection, making it straightforward for users to engage directly with the agent in teaching it their preferred methods and behaviors. Additionally, a method is required to use this data effectively, enabling agents to adapt to user-specific preferences and perform tasks in a manner that aligns with the user's individual standards and requirements. Together, these solutions would allow web navigation agents to be both effective and ethically aligned with the expectations of individual users.

Research Questions:

(1) How can we design a transparent and interactive framework for ethically collecting personalization data that lets users guide the learning process without compromising privacy?
(2) How can we develop a method to personalize web navigation agents using user-guided data, enabling them to adapt to individual user standards across diverse tasks and domains?
(3) How can we safeguard ethical and legal standards in personalized web agents, especially in cases where user preferences may introduce biases or inappropriate behaviors?

5 Methodology and Evaluation Approach

To address the limitations of current web navigation agents in providing ethical, user-guided personalization, I propose a framework that builds upon generative models with a personalized data collection process. This framework transitions from "default task execution" (standard generative model behavior) to "personalized task execution" based on explicit user guidance.

The agent operates in a partially observable web environment, where it performs two types of actions: **Interactive Actions** $\mathcal{A}$ and **Reasoning Actions** $\mathcal{L}$, similar to approach in [16]. Interactive Actions include operations like `click(elem)`, `type(elem, text)`, and tab management, which directly affect the environment, while Reasoning Actions represent internal LLM-generated thoughts that refine task planning. The action space $\hat{\mathcal{A}} = \mathcal{A} \cup \mathcal{L}$ enables the agent to navigate web pages effectively.

At each time step t, the agent receives an observation o_t and updates its context c_t as:

$$c_t = (o_1, a_1, \hat{a}_1, \ldots, o_{t-1}, a_{t-1}, \hat{a}_{t-1}, o_t),$$

where each o_i represents the visible state of the web page, and a_i, $\hat{a}_i$ denote interactive and reasoning actions. The agent's policy $\pi(a_t|c_t)$ generates a tuple $(a_t, \hat{a}_t)$ of an interactive action and a thought, which are selected based on the evolving context. For instance, during a form-filling task, the agent might produce a thought "Locate the username field" and select `click("username_field")`.

To enable personalization, I propose a **User-Guided Task Execution Framework**. This framework allows users to specify task preferences that differ from the default execution. The agent stores personalized task executions as a directed graph G_{user} where:

- **Nodes (States)** s_t represent observations of the environment.
- **Edges (Transitions)** represent actions and thoughts, forming a graph that maps out the user's preferred sequence of task steps.

Once a personalized execution is complete, a **memory stream** $M_{\text{user}} = \{(s_t, a_t^{\text{user}}, \hat{a}_t, r_t^{\text{user}})\}_{t=1}^{T}$ records the user's actions, thoughts, and any reasons for deviations from default execution. This information is saved in a **personalization bank** $\mathcal{B}_{\text{user}}$ as high-level preferences P_{user}, enabling the agent to generalize these insights across tasks.

The agent follows a layered approach for task execution. If a task-specific G_{user} exists, it guides actions; otherwise, a personalized policy $\pi_{\text{personalized}}(a_t|c_t, P_{\text{user}})$ derived from P_{user} is used. If no personalized information is available, the agent defaults to $\pi(a_t|c_t)$.

Evaluation Strategy: To evaluate the effectiveness of personalized task executions, I will use standard metrics like task success rate, completion time, and error rate. Additionally, I will conduct A/B testing to compare user satisfaction and alignment with user preferences between personalized and default executions. User studies will generate "User Satisfaction Scores" and "Personalization Alignment Scores" for more nuanced insights, addressing personalization's impact on user experience.

6 Research Issues for Discussion

1. Ethical, Transparent Data Collection for Personalization. This research introduces an interactive, privacy-conscious framework for personalization, emphasizing transparency in data collection. Unlike passive data collection, my approach limits interactions to structured, user-driven sessions, mitigating privacy concerns and avoiding reinforcement of undesirable habits. Discussion will focus on refining these practices to maintain user trust while achieving effective personalization.

2. Evaluating Personalized Task Executions in Web Navigation Agents. Current benchmarks, such as WebArena [20] and AgentBench [10], do not support personalization, presenting challenges in assessing agent performance on user-specific preferences. I am exploring new evaluation metrics, such as "Personalization Alignment Scores," and user study designs to capture the effectiveness and ethicality of personalized agent behavior. Discussions will center on best practices for evaluating and scaling personalized interactions, particularly in domains requiring high adaptability.

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